SECTION 8 – SHIPPING AND DELIVERY DATA

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8–1 SHIPMENT

Note

The Signa 1.5T Twin*Speed* system is shipped in 3 steps: 1) GE MDP and TSCC or TGWC equipment 2) Magnet with partial Enclosure and accessories 3) remaining system equipment. Refer to **Section 9–8 TYPICAL MR INSTALLATION PROJECT SCHEDULE** for more information.

Domestic transportation for the MR system, excluding the magnet, will be via an air–ride moving van. The magnet will be shipped on an air–ride flat–bed truck. Export transportation for the MR system overseas will be via air shipment in a pressurized cargo hold. See Table 8–2 for the shipping weights and dimensions of the major MR system components. Actual shipping may vary, international shipment may require equipment to be crated. Refer to Table 8–1 for transportation environmental conditions.

TABLE 8–1
TRANSPORTATION ENVIRONMENTAL CONDITIONS

SYSTEM EQUIPMENT	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	RELATIVE HUMIDITY %	HUMIDITY CHANGE (%/Hr)	ATMOSPHERIC PRESSURE hPa
Electronics Cabinets & equipment	–30 to 140 (–34 to 60)	176 (80)	0–90 non–condensing	30	1012 to 525
Magnet	–31 to 122 (–35 to 50)	176 (80)	0–90 non–condensing	30	1012 to 525

The GE magnet utilizes a Shield/Cryo Cooler System to maintain a reduced helium boil–off. However, the Shield Cooler System is not operational during transportation. Therefore the magnet will require liquid helium replenishment if transportation time exceeds two weeks. Contact GE Service for magnet servicing.

The LCC Magnet is filled with liquid helium at initial shipment but can be allowed to warm up during transportation without damage to the support structure.

8–2 STORAGE REQUIREMENTS

If the system is stored before installation, it must be stored in a warehouse protected from weather. The storage temperature should be between –30 and 140 degrees F (–34 and 60 degrees C) and the relative humidity between 0 and 90% (non–condensing).

There are two scenarios for storing the magnet. One is to maintain the magnet cold temperature by connecting and operating the Shield/Cryo Cooler System. Periodic replenishment of liquid helium may be required depending on the storage time. The other scenario is to store the magnet without maintaining the cold temperature. The LCC magnet can be moved cold or warm, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation* for magnet moving requirements and details. Contact GE Service for necessary servicing before moving the magnet.

8–3 MAGNET CONSIDERATIONS

For domestic, the magnet is shipped covered with plastic (no shipping pallet). For export, the magnet is crated for shipment on a special shipping pallet. Refer to Table 8–2 for the weight and dimensions of the magnet in its cold ship configuration (i.e. with liquid cryogens in vessel within the cryostat) and with the RF/Gradient Coil inside the magnet bore.

The magnet moving dimensions are shown in Illustration 8–1.

Consideration must be given to the delivery route of the magnet to ensure that the floor can support the magnet and any rigging equipment required to move it. A structural analysis should be performed by a professional structural engineer. The magnet must not be tilted more than 30° in any direction when being moved into position.

The customer is to provide and arrange for riggers to move the magnet from the delivery truck to the final site location. Contact local GE Service for a list of recommended rigger companies. The customer's riggers should have an adequate amount of liability insurance to cover any damage to property or MR system that may occur during delivery of the magnet. The GE sales representative or installation specialist can provide customer riggers with the replacement value of the MR system for insurance purpose.

8–4 COLD–SHIPPED MAGNET DELIVERIES

Cold–shipped magnets are those magnets which are shipped with liquid helium in the vessel within the cryostat. Thus, when these magnets arrive at site, a cryogen delivery route must be available for moving cryogen dewars to the magnet for periodic replenishment of liquid helium. Also, means must be provided for venting of the cryogenic gases if the GE supplied venting kit is not yet installed and if the RF shielded room vent opening is not completed.

Note

Power and cooling water for the shield cooler cabinet must be available when the magnet is delivered to minimize cryogens usage. See Section 4–4, WATER COOLING REQUIREMENTS, and Section 5, POWER REQUIREMENTS, for specifications.

TABLE 8–2
SHIPPING DATA

MR COMPONENT	APPROXIMATE W x D x H	APPROXI-MATE WEIGHT	METHOD OF SHIPMENT
LCC Magnet with cryogens, partial Quiet Technology Enclosure installed	93 x 144 x 107 (2362 x 3658 x 2718)	See Note 1	Domestic – Tarped International – crate/pallet
Magnet Accessory Equipment	48 x 48 x 28 (1219 x 1219 x 711)	400 (182)	crate
Shield/Cryo Cooler Compressor Cabinet	26 x 28 x 42 (660 x 711 x 1067)	240 (109)	skid with box cover
Rear Pedestal Assembly with Rear Split Bridge Assembly	24 x 48 x 45 (610 x 1219 x 1143)	60 (27)	box
Enclosure Top	48 x 36 x 36 (1219 x 914 x 914)	30 (14)	box
Enclosure Skirts	40 x 24 x 24 (1016 x 610 x 610)	30 (14)	box
Patient Table	95 x 36 x 50 (2413 x 914 x 1270)	670 (305)	pallet
Patient Blower Box	24 x 30 x 24 (610 x 762 x 610)	30 (14)	box
1.5T SRFD II Cabinet	24 x 38 x 72 (610 x 965 x 1829)	500 (225)	on cabinet casters, wrapped with plastic
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	2140 (970)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
System Control Cabinet for Signa Twin*Speed* with EXCITE Technology	24 x 33 x 78 (610 x 838 x 1981)	475 (215)	on cabinet casters, wrapped with plastic
System Control Cabinet for Signa Twin*Speed*	24 x 33 x 78 (610 x 838 x 1981)	500 (227)	on cabinet casters, wrapped with plastic
Twin Accessory Cabinet	23 x 39 x 50 (584 x 991 x 1270)	600 (272)	on cabinet casters, wrapped with plastic
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Operator Workspace Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	box
Operator Workspace LCD Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	skid
Operator Workspace equipment	32 x 32 x 23 (813 x 813 x 584)	100 (45)	box
Operator Workspace Table	45 x 54 x 37 (1143 x 1372 x 940)	180 (82)	box
MNS Cabinet *	24 x 42 x 71 (610 x 1067 x 1803)	430 (195)	on cabinet casters, wrapped with plastic
BrainWave Cabinet *	24 x 23 x 72 (610 x 584 x 1829)	350 (159)	on cabinet casters, wrapped with plastic
BrainWave Option Several Boxes *	40 x 48 x 32 (1016 x 1219 x 813)	165 (75)	all boxes on 1 pallet

Note * Optional Equipment

1 Approximate magnet shipping weight (includes packaging material) and weight of magnet with cryogens, TRM Gradient & RF coils, Enclosure parts installed on magnet, lifting beams, crate, and skid (i.e. minus packaging material): 12,195 lbs (5532 kg). International shipments must add shipping crate/pallet of 2,200 lbs (998 kg).

(Continued)

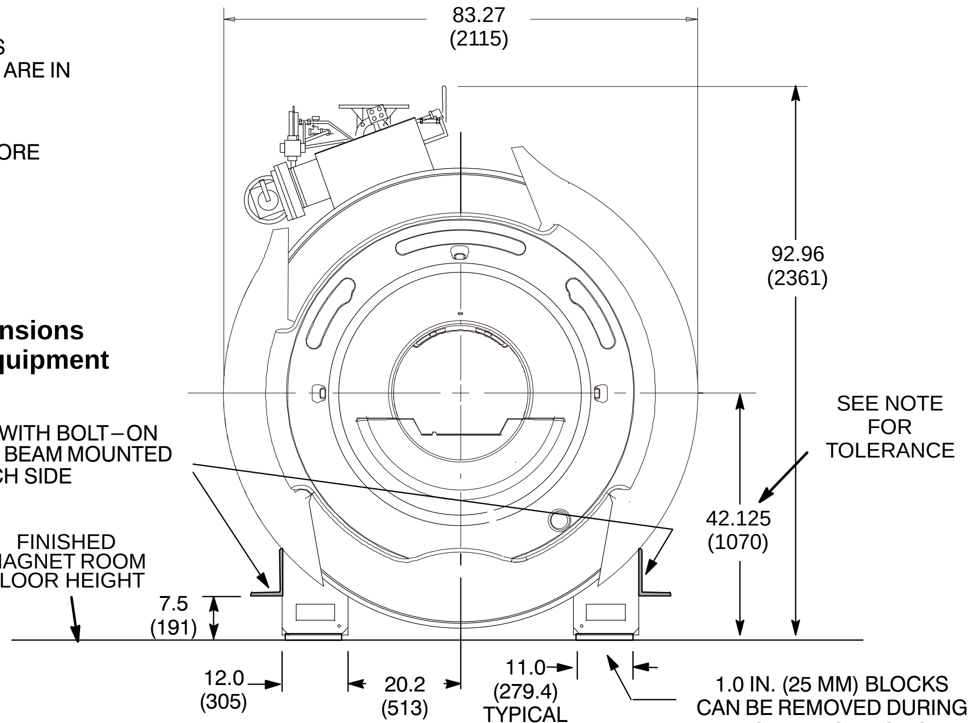
TABLE 8–2 (Continued)
SHIPPING DATA

MR COMPONENT	APPROXIMATE W x D x H	APPROXI-MATE WEIGHT	METHOD OF SHIPMENT
TSCC Outdoor Air Cooled equipment			
TSCC unit	52 x 82 x 75 (1321 x 2083 x 1905)	1320 (599)	crate/skid
Remote Control Panel	20 x 10 x 20 (51 x 25 x 51)	33 (15)	box
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	4 boxes
Ship loose items	32 x 32 x 32 (813 x 813 x 813)	50 (23)	box
TSCC Indoor Water Cooled equipment			
TSCC Indoor Water Cooled	58 x 38 x 82 (1473 x 953 x 2083)	1100 (499)	crate/skid
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	3 boxes
Ship loose items	32 x 32 x 32 (813 x 813 x 813)	50 (32)	box
TSCC Indoor Air Cooled equipment			
TSCC Indoor Air Cooled	68 x 36 x 80 (173 x 92 x 203)	1740 (789)	crate/skid
DowFrost HD glycol & ship loose items	30 x 50 x 70 (762 x 1270 x 1778)	300 (136)	box on pallet
TGWC Outdoor Air Cooled equipment			
TGWC unit	55 x 40 x 82 (141 x 102 x 208)	918 (417)	crate/skid
Remote Control Panel	13 x 7 x 15 (330 x 16 x 37)	20 (9)	box
DowFrost HD glycol & ship loose items	24 x 24 x 18 (610 x 610 x 457)	50 (23)	2 boxes
TGWC Indoor Water Cooled equipment			
TGWC Indoor Water Cooled	38 x 24 x 27 (97 x 61 x 69)	200 (91)	box on pallet
Remote Control Panel	12 x 12 x 12 (305 x 305 x 305)	20 (9)	box
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	2 boxes
Ship loose items	12 x 12 x 12 (305 x 305 x 305)	30 (14)	box
Ship loose items	26 x 6 x 6 (660 x 152 x 152)	15 (7)	box
TGWC Indoor Air Cooled equipment			
TGWC Indoor Air Cooled	34 x 40 x 76 (59 x 76 x 186)	918 (417)	crate/skid
DowFrost HD glycol & ship loose items	24 x 24 x 12 (610 x 610 x 305)	50 (23)	box

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MAGNET WITH CRYOGENS, TRM GRADIENT & RF COILS IN BORE WEIGHT, AND LIFTING BEAMS: 12,195 lbs (5532 kg)

CAUTION

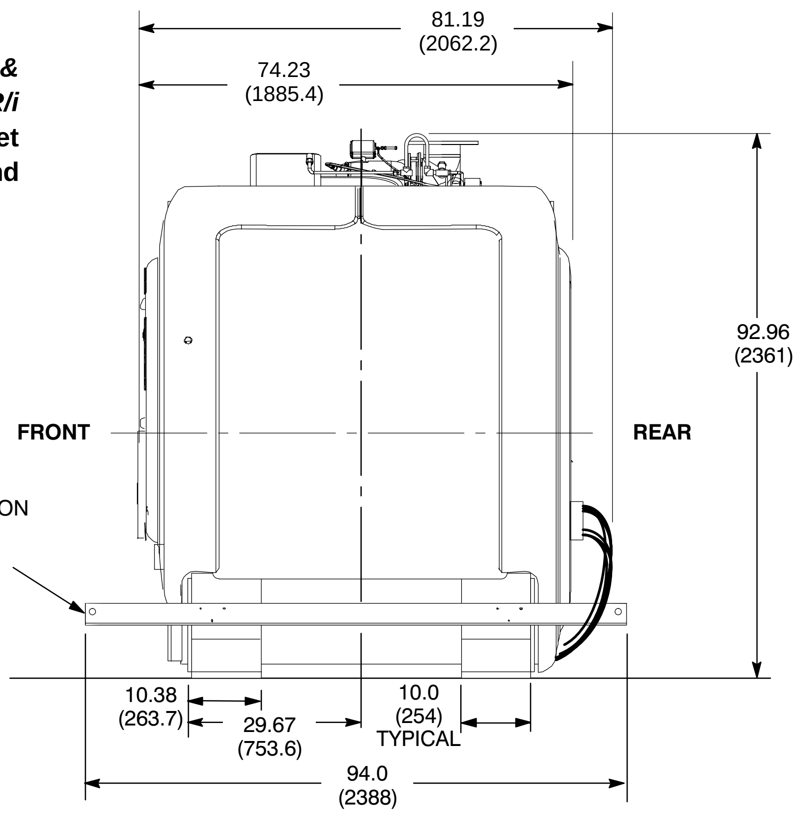
Final magnet moving dimensions are dependent on rigger equipment requirements.



FRONT VIEW

CAUTION

Refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation* for magnet moving and lifting requirements and details.



1.5T LCC (ACTIVE SHIELD) MAGNET MOVING DIMENSIONS
ILLUSTRATION 8–1